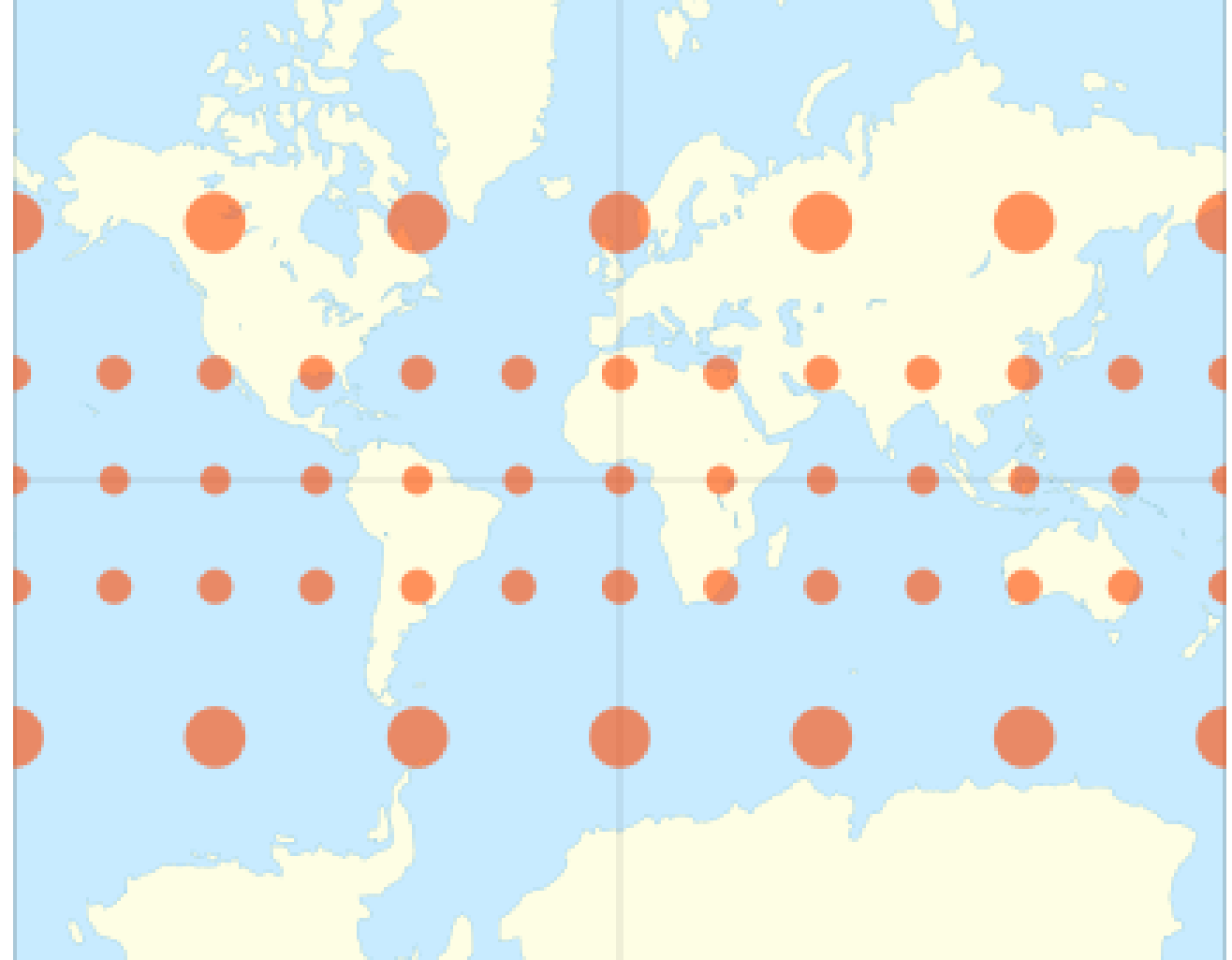
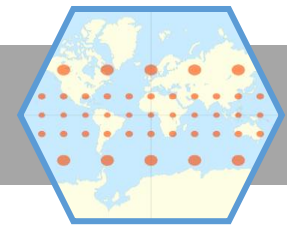


LAB 4: Cartography





EX 1: modulus of linear deformation

Plotting the modulus of linear deformation, considering these cases:

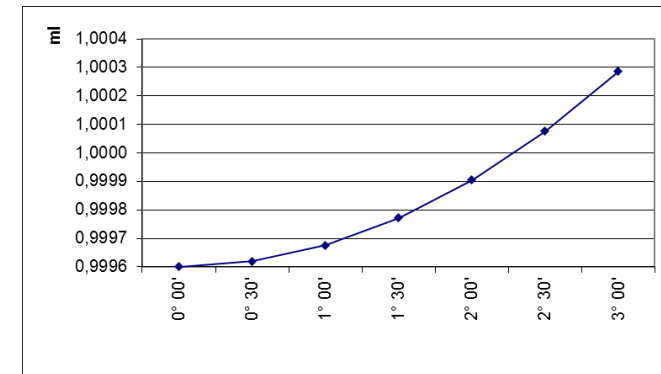
$$m_l = 0,9996 \left(1 + \frac{\lambda^2}{2} \cos^2 \varphi \right)$$

Case 1: lat= 30° long': 0°- 3°, with 0,5° of resolution

Case 2: lat= 37° long': 0°- 3°, with 0,5° of resolution

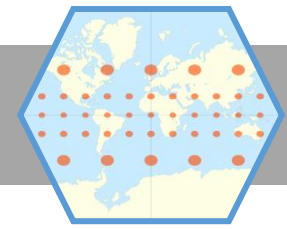
Case 3: lat= 45° long': 0°- 3°, with 0,5° of resolution

Case 4: lat= 60° long': 0°- 3°, with 0,5° of resolution



To compare the results.





EX 1: application of modulus of linear deformation

1) Define the coordinate (E, N) of POLITECNICO (DUCA site) and POLITECNICO (VALENTINO site) USING GOOGLE EARTH. You can select one point as «centroid» of the building.

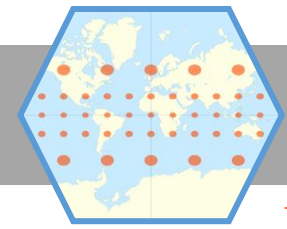
2) Calculate the Euclidian distance as Cartographic Distance

3) Calculate the real distance, considering the modulus of linear deformation, considering an average point in the middle

4) Make a comparison of the distances

$$m_l = 0,9996 \left(1 + \frac{\lambda^2}{2} \cos^2 \varphi \right)$$





EX 2: HIRVONEN equation

Using the Hirvonen equation, to transform these coordinates from geographic to cartographic.

P1 → lat = 45° 3' 45,717'' long = 7° 47' 26,292''

SR = WGS84 - UTM

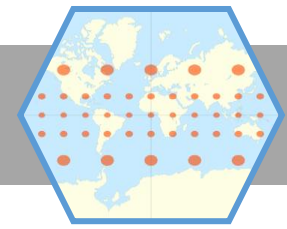
Central meridian = 9° UTM-ZONE= 32

P2 → lat = 38° 32' 34,649'' long = 16° 50' 06,493''

SR = WGS84 - UTM

Central meridian = 15° UTM-ZONE= 33





EX 2: HIRVONEN equation

$$x = R_P \operatorname{arcsenh} \frac{\cos \xi \tan \lambda'}{v}$$

$$y = a (A_1 \xi - A_2 \sin 2\xi + A_4 \sin 4\xi - A_6 \sin 6\xi)$$

where

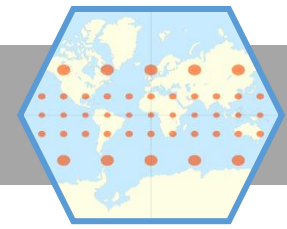
$$R_P = \frac{a^2}{c} \quad \text{Radius of polar curvature}$$

$$\lambda' = \lambda - \lambda_{mc} \quad \xi = \operatorname{arctg} \frac{\tan \varphi}{\cos(v_1 \lambda')}$$

$$v_1 = \sqrt{(1 + e'^2 \cos^2 \varphi)} \quad v = \sqrt{(1 + e'^2 \cos^2 \xi)}$$

$$\lambda' = (\lambda_P - \lambda_{mc}) \quad \text{Where } \lambda_{mc} \text{ is the central meridian of the fuse}$$





EX 2: HIRVONEN equation

$$A_1 = 1 - \frac{e^2}{4} - \frac{3e^4}{64} - \frac{5e^6}{256}$$

$$A_2 = \frac{3e^2}{8} + \frac{3e^4}{32} + \frac{45e^6}{1024}$$

$$A_4 = \frac{15e^4}{256} + \frac{45e^6}{1024}$$

$$A_6 = \frac{35e^6}{3072}$$

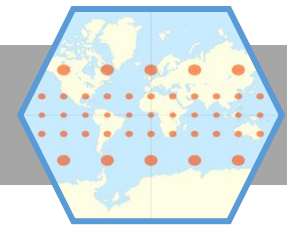
East = $x \cdot m_c$ + False East

North = $y \cdot m_c$

where m_c = contraction modulus 0,9996

*False East = depend for RS





EX 3: HIRVONEN equation (inverse)

Using the Hirvonen equation, to transform these coordinates from cartographic to geographic.

P → **East** = 470139,66 m **North** = 5031468,37 m

SR = WGS84 – UTM-ZONE 32

Long = $8^{\circ} 37' 5'',6209$

Lat = $45^{\circ} 26' 9'',9617$





EX 4: Copernicus images

Copernicus BROWSER EN ▼ Marco Piras ▼

VISUALISE SEARCH

SH DASHBOARD WORKSPACE

DATE: SINGLE

< YYYY-MM-DD > 30%

Show latest date ↗

Find products for current view

CONFIGURATIONS: Default ▼

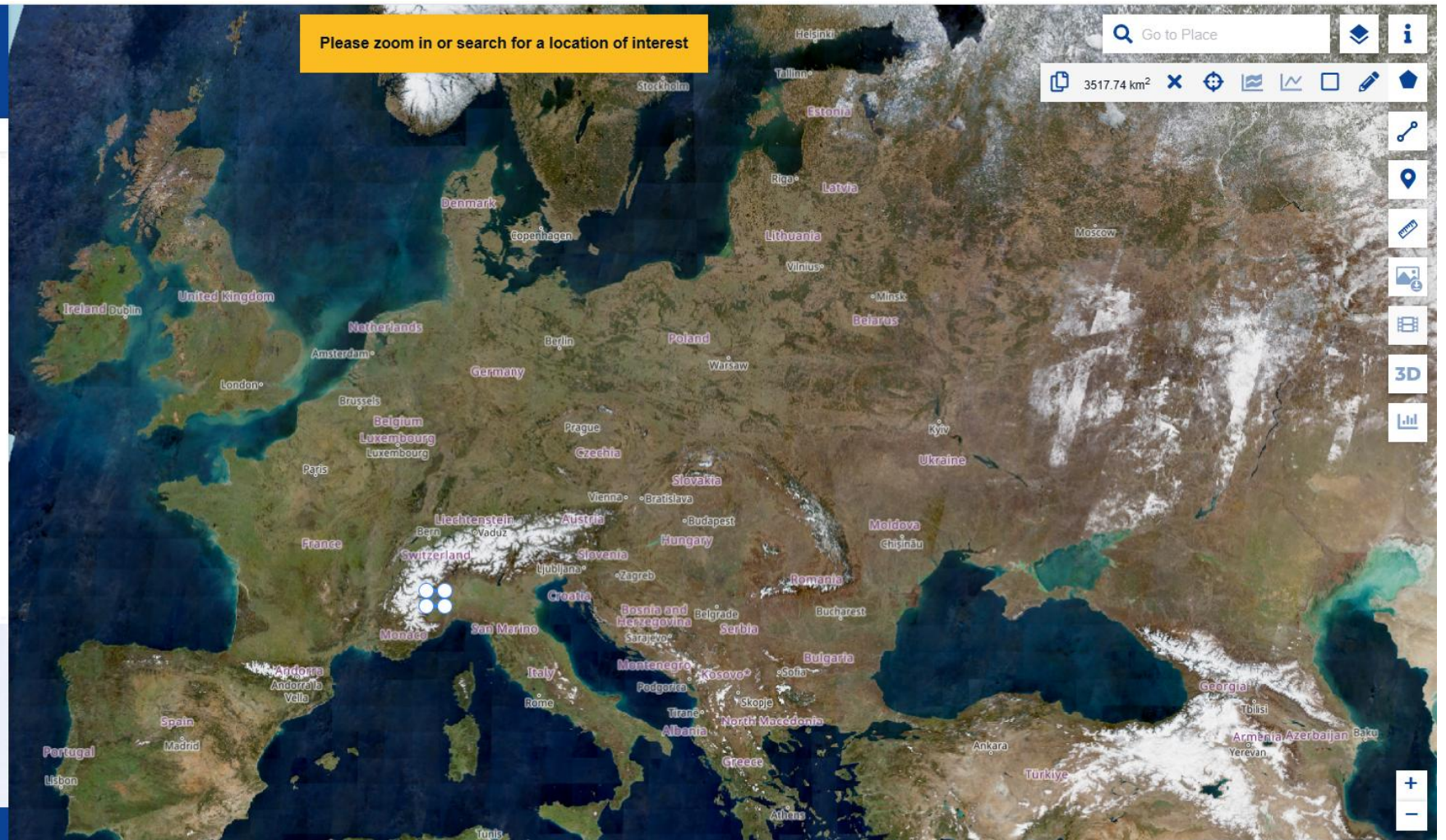
DATA COLLECTIONS:

Sentinel-2 ▼

Sentinel-2 L1C

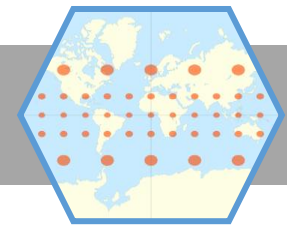
Sentinel-2 L2A ✓

European Union Copernicus ESA About Support



<https://browser.dataspace.copernicus.eu/>



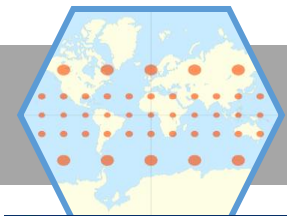


EX 4: Copernicus images

- **To Create a AOI and export the following products in two different year (2025 and 2020)**
 - **NDVI**
 - **Air quality (CO2 or Methane)**
 - **True color**
 - **False Color**

<https://browser.dataspace.copernicus.eu/>





EX 4: Copernicus images

Copernicus
BROWSER

EN ▾ Marco Piras ▾ <

VISUALISE

SEARCH

SH DASHBOARD

WORKSPACE

< 2025-05-14 >

☁ 30%

↗

▼

Change Detection through... ▾

▼

Sentinel-2 L2A ⓘ

🗲

★

↔

📌

▼

LAYERS:

True color
Based on bands 4,3,2

+ Add to </> ▾

False color
Based on bands 8,4,3

NDVI
Based on combination of bands (B8 - B4)/(B8 + B4)

Custom
Create custom visualisation

Show effects and advanced options

Hide layer

Share

🇪🇺 Copernicus
Sentinel Hub on Earth

eesa

About

Support

v1.10.2

Basic Analytical High-res print

Download

X

Image download

Show captions ⓘ ☒

Add map overlays ⓘ ☐

Add OSM background ⓘ ☐

Show legend ⓘ ☐

Crop to AOI ⓘ ☒

Draw AOI or Line on image ⓘ ☐

Description ⓘ

Image format: JPG (no georeference) ▾

Preview

Go to Place

3701.97 km²

✕

📏

📐

📊

📋

📌

📄

📅

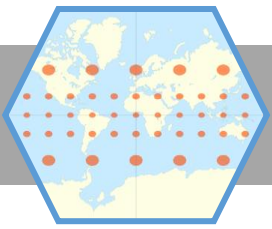
3D

📈

+

Lat: 44.8902, Lng: 11.3873

2 km



EX 4: Copernicus images

Basic

Analytical

High-res print

Download

×

Image download

Image format: TIFF (32-bit float)

Image resolution: HIGH
2500 x 477 px

Coordinate system: WGS 84 (EPSG:4326)
Resolution:
lat.: 0.0011065 deg/px (4.0sec/px)
long.: 0.0015656 deg/px (5.6sec/px)

Clip extra bands

Layers:

Visualised

Raw

☒ True color

☐ False color

☐ NDVI

☐ B01

☐ B02

☐ B03

☐ B04

☐ B05

☐ B06

☐ B07

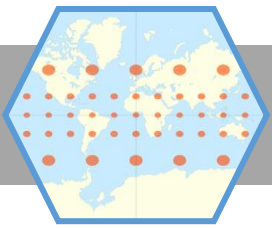
☐ B08

☐ B8A

☐ B09

Show more

Preview



EX 4: Copernicus images

- To create a TIMELASPE considering at least 6 years, about 2 different themes.

Timelapse

2025-04-03 - 2025-05-14

Filter by months

Select 1 image per:

☐ orbit ☒ day ☐ week ☐ month ☐ year

Sentinel-2 L2A: True color

[Search](#)

[Add layers from pins](#)

Visualisations

Min. tile coverage: 0%

Max. cloud coverage: 39%

☒ Select All

5 km

2025-04-23

Copernicus
Data Space Ecosystem

Speed: 1 fps Transition: None

6 / 8

[Download](#)